

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5331

5 Credits: 4

Program core

Source-grounded

Industrial Automation Control

□ To introduce the concepts of Process and Process control.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5331 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5331.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automation and Robotics
Course code	5331
Course title	Industrial Automation Control
Semester	5 Credits: 4
Course category	Program core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To introduce the concepts of Process and Process control.
- To illustrates various elements in a process control loop and controller tuning.
- Familiarize the basic concepts of PLC programming
- To explain functional elements of process control loop and various communication methods in process control applications.

Course outcomes

CO	Official outcome	Study level
CO1	Understand the concept of Process control and the 15 Understanding process control loops	See official syllabus
CO2	To understand the various control modes and to 16 Applying implement pneumatic and electronic controllers	See official syllabus
CO3	Familiarize the basic concepts of PLC programming 16 Understanding To understand telemetry system and Digital	See official syllabus
CO4	13 Applying Communication channels in Process Control	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 2
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Understand the concept of Process control and the process control loops	4	As specified
Module 2	and electronic controllers	4	As specified
Module 3	Familiarize basic concept of PLC programming	6	As specified
Module 4	Process Control	3	As specified

MODULE 1

Understand the concept of Process control and the process control loops

This module is presented from the official syllabus scope for Industrial Automation Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the concept of Process control and the process control loops
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Define the concepts of process control.

Define the concepts of process control. is an official topic in Module 1 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define the concepts of process control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define the concepts of process control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define the concepts of process control. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Define the concepts of process control.
 റിസോൺസ് ചെയ്ത് definition/meaning നൽകുക. റിസോൺസ് ചെയ്ത്
 റിസോൺസ് ചെയ്ത്, steps, application നൽകുക. Technical terms
 English-ൽ നൽകുക.

▼ Explain automatic control 3 Understanding Explain the process and controller

Explain automatic control 3 Understanding Explain the process and controller is an official topic in Module 1 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain automatic control 3 Understanding Explain the process and controller using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain automatic control 3 understanding explain the process and controller should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain automatic control 3 Understanding Explain the process and controller means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain automatic control 3 Understanding Explain the process and controller definition/meaning . . . , steps, application Technical terms English-

▼ 4 Understanding Characteristics

4 Understanding Characteristics is an official topic in Module 1 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Characteristics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding characteristics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Characteristics means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding Characteristics
 പദപരിചയം definition/meaning. പദപരിചയം, steps, application പദപരിചയം പദപരിചയം. Technical terms
 English- പദപരിചയം പദപരിചയം.

▼ Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag -Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic

Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag -Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic is an official topic in Module 1 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error,

set point, controlled variable, manipulated variable and measured variable- Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag -Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the different process control loops 5 understanding automatic control-block diagram of process control - error, set point, controlled variable, manipulated variable and measured variable-elements of process control loop-process characteristics - process equation, process load, process lag and self regulation, control system parameters - error, variable range, control parameter range, control lag -dead time, cycling-process parameters-temperature process control, pressure process control system-flow process control system, level process control system. to understand the various control modes and to implement pneumatic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable- Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag -Dead Time, Cycling-Process Parameters-Temperature process

Aspect	What to prepare
	control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag - Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic definition/meaning. steps, application. Technical terms English-.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

and electronic controllers

This module is presented from the official syllabus scope for Industrial Automation Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and electronic controllers
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Describe different types of process 4 Understanding To understand the various control modes

Describe different types of process 4 Understanding To understand the various control modes is an official topic in Module 2 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe different types of process 4 Understanding To understand the various control modes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe different types of process 4 understanding to understand the various control modes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Describe different types of process 4 Understanding To understand the various control modes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Describe different types of process 4 Understanding To understand the various control modes definition/meaning . steps, application . Technical terms English- .

▼ 6 Understanding continuous and discontinuous Understand the implementation of Electronic

6 Understanding continuous and discontinuous Understand the implementation of Electronic is an official topic in Module 2 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding continuous and discontinuous Understand the implementation of Electronic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding continuous and discontinuous understand the implementation of electronic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding continuous and discontinuous Understand the implementation of Electronic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 6 Understanding continuous and discontinuous Understand the implementation of Electronic definition/meaning steps, application Technical terms English-

▼ 3 Understanding Controllers

3 Understanding Controllers is an official topic in Module 2 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Controllers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding controllers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Controllers means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding Controllers

നിയന്ത്രണകർമ്മങ്ങൾക്ക് നിർവ്വചന/അർത്ഥം നൽകുക. നിർവ്വചനം നൽകുന്നതിനുള്ള ഘട്ടങ്ങൾ, പ്രയോഗം നൽകുന്നതിനുള്ള നിർവ്വചനം. Technical terms English-ൽ നൽകുക.

▼ Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode - proportional band and offset error - integral control Mode - reset rate - derivative control Mode - derivative time- composite control Modes- PI, PD and PID control Modes-Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods- open loop method-Ziegler Nichols method.

Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode - proportional band and offset error - integral control Mode - reset rate - derivative control Mode - derivative time- composite control Modes- PI, PD and PID control Modes-Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods- open loop method-Ziegler Nichols method. is an official topic in Module 2 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Tuning of controllers 3 Applying neutralzone - multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode - reset rate - derivative control Mode - derivative time- composite control Modes- PI, PD and PID control Modes-Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning

methods- open loop method-Ziegler Nichols method. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate tuning of controllers 3 applying neutralzone - multi position control mode- continuous control modes - proportional control mode -proportional band and offset error - integral control mode - reset rate - derivative control mode - derivative time- composite control modes- pi, pd and pid control modes-electronic controllers-electronic error detector-electronic p,pi,pd and pid controller using op -amp-tuning methods- open loop method- ziegler nichols method. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional control Mode -proportional band and offset error - integral control Mode - reset rate - derivative control Mode - derivative time- composite control Modes- PI, PD and PID control Modes-Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods- open loop method-Ziegler Nichols method. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇല്ലാത്ത Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode - reset rate - derivative control Mode - derivative time-composite control Modes- PI, PD and PID control Modes-Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods- open loop method-Ziegler Nichols method.
 ഇതിന്റെ definition/meaning എന്താണ്.
 steps, application
 Technical terms English-ൽ എഴുതുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Familiarize basic concept of PLC programming

This module is presented from the official syllabus scope for Industrial Automation Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Familiarize basic concept of PLC programming
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and

Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand sequence and logic control 1 understanding describe a programmable logic controller and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and definition/meaning . steps, application . Technical terms English- .

▼ 2 Understanding its components

2 Understanding its components is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding its components using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding its components should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding its components means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding its components
 പഠിക്കേണ്ടതെന്തെന്ന് definition/meaning പറയണം. പഠിക്കേണ്ട ഘട്ടങ്ങൾ, steps, application പറയണം. Technical terms English-ൽ പറയണം.

▼ Understand Ladder diagram

Understand Ladder diagram is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Ladder diagram using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand ladder diagram should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Ladder diagram means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- ഉറപ്പ് Understand Ladder diagram

ഉറപ്പ് definition/meaning ഉറപ്പ്. ഉറപ്പ് ഉറപ്പ്, steps, application ഉറപ്പ് ഉറപ്പ്. Technical terms English-ഉറപ്പ് ഉറപ്പ്.

▼ PLC programming

PLC programming is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify PLC programming using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, plc programming should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What PLC programming means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- PLC programming പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English- പദപരിചയം പദപരിചയം.

▼ Case studies on PLC Ladder programming 3

Case studies on PLC Ladder programming 3 is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Case studies on PLC Ladder programming 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, case studies on plc ladder programming 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Case studies on PLC Ladder programming 3 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Case studies on PLC Ladder programming 3 definition/meaning. steps, application Technical terms English-

▼ Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in

Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand various motion controllers 3 elements, ladder diagram, plc, programming of plc- analog and digital i/os, timers, counters, function blocks. case studies on plc ladder programming. motion controllers- vfd, mld, external relays and contactors. to understand telemetry system and digital communication channels in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming.

Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Process Control

This module is presented from the official syllabus scope for Industrial Automation Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Process Control
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus

Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus is an official topic in Module 4 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the different telemetry systems 3 understanding explain the concept of foundation field bus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus definition/meaning . steps, application . Technical terms English- .

▼ 6 and Profit bus DP Explain the concept of HART Communication

6 and Profit bus DP Explain the concept of HART Communication is an official topic in Module 4 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 and Profit bus DP Explain the concept of HART Communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 and profit bus dp explain the concept of hart communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 6 and Profit bus DP Explain the concept of HART Communication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-൬ 6 and Profit bus DP Explain the concept of HART Communication ുറക്കുക definition/meaning ുറക്കുക. ുറക്കുക ുറക്കുക ുറക്കുക, steps, application ുറക്കുക ുറക്കുക. Technical terms English-൬ ുറക്കുക ുറക്കുക.

▼ **3 Applying Protocol System- Current Telemetry System - Motion balance and Force balance telemetry systems - Position Telemetry System- Digital communication channels-Fieldbus- advantages of fieldbus-ProfibusDP- Function elements in Foundation Fieldbus.Highway - Addressable Remote Transducer-Benefits of HART field communication protocol. Working of HART - Block diagram of HART digital communication System - HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology, PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILEY India. Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/108105088> 2 https://onlinecourses.nptel.ac.in/noc21_me67/preview 3 <https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf> 4 <https://processcontrol.tv/tutorials> 5 <https://www.pidlab.com/en/virtual-labs>**

3 Applying Protocol System- Current Telemetry System - Motion balance and Force balance telemetry systems - Position Telemetry System- Digital communication channels-Fieldbus- advantages of fieldbus-ProfibusDP- Function elements in Foundation Fieldbus.Highway - Addressable Remote Transducer-Benefits of HART field communication protocol. Working of HART - Block diagram of HART digital communication System - HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology, PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4

Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P.

Eckman, Automatic process control, WILLEY India. Online Resources Sl.No

WebsiteLink 1 <https://nptel.ac.in/courses/108105088> 2

https://onlinecourses.nptel.ac.in/noc21_me67/preview 3

<https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf> 4

<https://processcontrol.tv/tutorials> 5 <https://www.pidlab.com/en/virtual-labs>

is an official topic in Module 4 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying Protocol System- Current Telemetry System - Motion balance and Force balance telemetry systems - Position Telemetry System- Digital communication channels-Fieldbus- advantages of fieldbus- Profibus DP- Functional elements in Foundation Fieldbus Highway - Addressable Remote Transducer- Benefits of HART field communication protocol. Working of HART - Block diagram of HART digital communication System - HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology, PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No WebsiteLink 1 <https://nptel.ac.in/courses/108105088> 2 https://onlinecourses.nptel.ac.in/noc21_me67/preview 3 <https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf> 4

<https://processcontrol.tv/tutorials> 5 <https://www.pidlab.com/en/virtual-labs> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying protocol system- current telemetry system - motion balance and force balance telemetry systems - position telemetry system- digital communication channels- fieldbus- advantages of fieldbus- profibus dp- functional elements in foundation fieldbus. highway - addressable remote transducer- benefits of hart field communication protocol. working of hart - block diagram of hart digital communication system - hart specification (layers of hart). text / reference t/r book title/author industrial instrumentation, control and automation, s. mukhopadhyay, s. sen and t1 a. k. deb, jaico publishing house, 2013 r1 jonathan love, process automation handbook, springer 2007 r2 curtis.d. johnson, process control instrumentation technology, phi. r3 liptak, volume ii, instrument engineers hand book, chilton book company. r4 krishnakanth, computer based industrial control, phi. r5 donald.p. eckman, automatic process control, willey india. online resources sl.no websitelink 1 <https://nptel.ac.in/courses/108105088> 2 https://onlinecourses.nptel.ac.in/noc21_me67/preview 3 <https://scienceq.org/wp-content/uploads/2018/10/jaetv7i102.pdf> 4 <https://processcontrol.tv/tutorials> 5 <https://www.pidlab.com/en/virtual-labs> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying Protocol System- Current Telemetry System - Motion balance and Force balance telemetry systems - Position Telemetry System- Digital communication channels- Fieldbus- advantages of fieldbus- Profibus DP- Functional elements in Foundation Fieldbus. Highway - Addressable Remote Transducer- Benefits of HART field communication protocol. Working of HART - Block diagram of HART digital

Aspect	What to prepare
	communication System - HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology, PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILEY India. Online Resources Sl.No WebsiteLink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67/preview 3 https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Applying Protocol System- Current Telemetry System - Motion balance and Force balance telemetry systems - Position Telemetry System- Digital communication channels-Fieldbus- advantages of fieldbus-Profibus DP- Functionalelements in Foundation Fieldbus.Highway - Addressable Remote Transducer-Benefitsof HART field communication protocol. Working of HART - Block diagram of HART digital communication System - HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S.

Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology, PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILEY India. Online Resources Sl.No WebsiteLink 1

<https://nptel.ac.in/courses/108105088> 2

https://onlinecourses.nptel.ac.in/noc21_me67/preview 3

<https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf> 4

<https://processcontrol.tv/tutorials> 5 <https://www.pidlab.com/en/virtual-labs>

പ്രക്രിയ നിയന്ത്രണം നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുന്നു. നിർവ്വചനം നിർവ്വചിക്കുന്നു

നിർവ്വചിക്കുന്നു, steps, application നിർവ്വചിക്കുന്നു നിർവ്വചിക്കുന്നു. Technical terms

English-ൽ നിർവ്വചിക്കുന്നു നിർവ്വചിക്കുന്നു.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Define the concepts of process control.. (3 marks)**

► **Define or explain Explain automatic control 3 Understanding Explain the process and controller. (3 marks)**

► **Define or explain 4 Understanding Characteristics. (3 marks)**

► **Define or explain Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag -Dead Time, Cycling-Process Parameters- Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic. (3 marks)**

Module 2

► **Define or explain Describe different types of process 4 Understanding To understand the various control modes. (3 marks)**

► **Define or explain 6 Understanding continuous and discontinuous**

Understand the implementation of Electronic. (3 marks)

► **Define or explain 3 Understanding Controllers. (3 marks)**

► **Define or explain Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional control Mode -proportional band and offset error - integral control Mode - reset rate - derivative control Mode - derivative time-composite control Modes- PI, PD and PID control Modes-Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods- open loop method-Ziegler Nichols method.. (3 marks)**

Module 3

► **Define or explain Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and. (3 marks)**

► **Define or explain 2 Understanding its components. (3 marks)**

► **Define or explain Understand Ladder diagram. (3 marks)**

► **Define or explain PLC programming. (3 marks)**

Module 4

► **Define or explain Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus. (3 marks)**

► **Define or explain 6 and Profit bus DP Explain the concept of HART Communication. (3 marks)**

► **Define or explain 3 Applying Protocol System- Current Telemetry System - Motion balance and Force balance telemetry systems - Position Telemetry System- Digital communication channels-Fieldbus- advantages of fieldbus- Profibus DP- Function elements in Foundation Fieldbus. Highway - Addressable Remote Transducer- Benefits of HART field communication protocol. Working of HART - Block diagram of HART digital communication System - HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology, PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/108105088> 2 https://onlinecourses.nptel.ac.in/noc21_me67/preview 3 <https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf> 4 <https://processcontrol.tv/tutorials> 5 <https://www.pidlab.com/en/virtual-labs>. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define the concepts of process control.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe different types of process 4 Understanding To understand the various control modes.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and.**

Module 4

1-mark definition: State the meaning of **Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No WebsiteLink <https://nptel.ac.in/courses/108105088>
- https://onlinecourses.nptel.ac.in/noc21_me67 /preview
- <https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf>
<https://processcontrol.tv/tutorials> <https://www.pidlab.com/en/virtual-labs>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5331. Verify institutional updates against the current official curriculum.